1 Halfspace

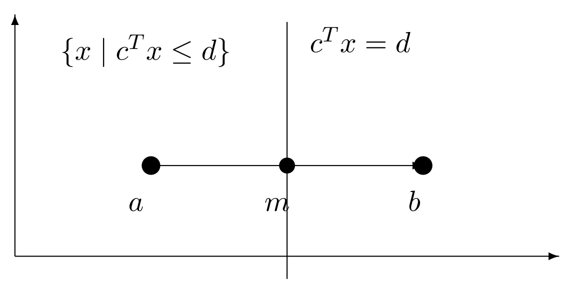
$$\|x-a\|^2 \leq \|x-b\|^2$$
$$(x-a)^T(x-a) \leq (x-b)^T(x-b)$$
$$x^Tx - 2a^Tx + a^Ta \leq x^Tx - 2b^Tx + b^Tb$$

Canceling x^Tx and rearranging gives

$$2(b-a)^Tx \leq b^Tb - a^Ta$$
$$(b-a)^Tx \leq \frac{\|b\|^2 - \|a\|^2}{2}$$

So $c = b - a$ and $d = \frac{\|b\|^2 - \|a\|^2}{2}$. The boundary hyperplane is perpendicular to the segment from a to b and passes through $m = \frac{a+b}{2}$.

The halfspace $c^Tx \leq d$ is the left side containing a



2 Proof of Cauchy-Schwarz Inequality

(a)

We need to check that its minimum value is also nonnegative. The quadratic is minimized at

$$\lambda^\star = -\frac{b}{c}$$

Substituting gives

$$a + 2b\left(-\frac{b}{c}\right) + c\left(-\frac{b}{c}\right)^2 = a - \frac{b^2}{c} \geq 0 \quad \implies b^2 \leq ac \quad \implies |b| \leq \sqrt{ac}$$

(b)

The inner product of any vector with itself is the sum of the squares of its entries, so it is always nonnegative:

$$(v + \lambda w)^T(v + \lambda w) = \|v + \lambda w\|^2 \geq 0$$

(c)

$$(v + \lambda w)^T(v + \lambda w) = v^T v + 2(v^T w)\lambda + (w^T w)\lambda^2$$

This has the same form as part (a), with

$$a = v^T v, \qquad b = v^T w, \qquad c = w^T w$$

So

$$|v^T w| \leq \sqrt{(v^T v)(w^T w)} = \sqrt{v^T v}\sqrt{w^T w}$$

(d)

Equality holds iff v and w are linearly dependent

For example, with $v = [1 \quad 0]^\top$ and $v = [2 \quad 0]^\top$:

$$|v^T w| = \sqrt{v^T v}\sqrt{w^T w}$$
$$2 = 1 \cdot 2 = 2$$

3 Right Inverses

For

$$A = \begin{bmatrix} -1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

we first solve $Az = y$. With $z = [z_1, \dots, z_5]^\top$, the equations are

$$-z_1 - z_4 + z_5 = y_1, \qquad z_2 + z_3 = y_2, \qquad z_1 + z_4 = y_3.$$

So z_1, z_2, and z_4 are free, with

$$z_3 = y_2 - z_2, \qquad z_5 = y_1 + y_3.$$

Applying this to the standard basis vectors $y = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, every right inverse has the form

$$B = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ -b_1 & 1 - b_2 & -b_3 \\ -a_1 & -a_2 & 1 - a_3 \\ 1 & 0 & 1 \end{bmatrix}$$

where $a_i, b_i \in \mathbb{R}$. For any such choice, $AB = I$.

(a) Yes

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

```
julia> B = [0 0 0;
            0 0 0;
            0 1 0;
            0 0 1;
            1 0 1];

julia> A = [-1 0 0 -1 1;
            0 1 1 0 0;
            1 0 0 1 0];

julia> A*B
3×3 Matrix{Int64}:
 1  0  0
 0  1  0
 0  0  1
```

(b) No. If $AB = I$ and $Bx = 0$, then

$$x = Ix = ABx = A0 = 0.$$

So $\text{null}(B) = \{0\}$, which has dimension zero, not one.

(c) No. If the third column of B were zero, then the third column of AB would also be zero. But $AB = I$, so the third column must be $\mathbf{e}_3 \neq 0$.

(d) Yes

$$B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

```
julia> B = [0 0 0;
            0 1//2 0;
            0 1//2 0;
            0 0 1;
            1 0 1];

julia> A*B
3×3 Matrix{Rational{Int64}}:
 1  0  0
 0  1  0
 0  0  1
```

(e) No. Every right inverse has fifth row $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, so $B_{51} = B_{53} = 0$, is impossible.

(f) Yes. The matrix from part (a).

4 Some True/False Questions

(a) False. Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Both have nonnegative entries and both are onto, but

$$A + B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

has rank one, so it is not onto.

(b) True. If

$$Ax = 0, \qquad (A+B)x = 0, \qquad (A+B+C)x = 0,$$

then $Bx = 0$ follows from the first two equations, and $Cx = 0$ follows from all three. Thus $x \in \text{null}(A) \cap \text{null}(B) \cap \text{null}(C)$.

(c) False. As a counterexample, taake $A = 0$, $B = 1$, and $C = 1$. Then

$$\text{null}\left(\begin{bmatrix} A \\ AB \\ ABC \end{bmatrix}\right) = \mathbf{R},$$

but

$$\text{null}(A) \cap \text{null}(B) \cap \text{null}(C) = \{0\}.$$

(d) True. For any x,

$$x^T(B^T A^T AB + B^T B)x = \|ABx\|^2 + \|Bx\|^2.$$

This is zero if and only if $Bx = 0$. Hence the nullspace is $\text{null}(B)$.

(e) True. The rank of a block diagonal matrix is the sum of the ranks of its diagonal blocks:

$$\text{rank} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} = \text{rank}(A) + \text{rank}(B).$$

If the block diagonal matrix is full rank, then neither diagonal block can be rank deficient. Therefore both A and B are full rank.

(f) True. The matrix $\begin{bmatrix} A & 0 \end{bmatrix}$ has the same column space as A. If $\begin{bmatrix} A & 0 \end{bmatrix}$ is onto, then A is onto, so A is full rank.

(g) True. Since A^2 is defined, A is square. If A^2 is onto, then for every y there is some x such that

$$A^2 x = y.$$

Let $z = Ax$. Then

$$Az = y.$$

So every y is in the range of A, which means A is onto.

(h) False. Let

$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then $A^T A = [1]$, which is onto as a map $\mathbf{R} \to \mathbf{R}$. But $A : \mathbf{R} \to \mathbf{R}^2$ is not onto.

(i) True. Suppose $\alpha_i \geq 0$ and

$$\sum_{i=1}^{k} \alpha_i u_i = 0.$$

Then

$$0 = \left\| \sum_{i=1}^{k} \alpha_i u_i \right\|^2 = \sum_{i=1}^{k} \sum_{j=1}^{k} \alpha_i \alpha_j u_i^T u_j.$$

Every term is nonnegative, and the diagonal terms are $\alpha_i^2 \|u_i\|^2$. Since each $u_i \neq 0$, all α_i must be zero.

(j) True. If

$$Ax + By = 0,$$

then $Ax = -By$. The vector Ax lies in the column space of A, and By lies in the column space of B. Since $A^T B = 0$, these two column spaces are orthogonal. Hence

$$0 = \|Ax + By\|^2 = \|Ax\|^2 + \|By\|^2,$$

so $Ax = 0$ and $By = 0$. Because A and B are skinny and full rank, $x = 0$ and $y = 0$. Therefore $[A\ B]$ has independent columns, so it is skinny and full rank.

5 Transmit Powers in a Wireless Network

(a)

Let

$$D = \operatorname{diag}(G_{11}, \ldots, G_{nn}), \qquad K = G - D.$$

Here Dp is the vector of received signal powers, and Kp is the vector of interference powers. The condition $S_i = S^{\text{target}}$ for every i is

$$G_{ii}p_i = S^{\text{target}} \left(\sigma + \sum_{j \neq i} G_{ij}p_j \right).$$

In vector form this is

$$Dp = S^{\text{target}}\sigma\mathbf{1} + S^{\text{target}}Kp,$$

or

$$(D - S^{\text{target}}K)p = S^{\text{target}}\sigma\mathbf{1}.$$

Assuming $D - S^{\text{target}}K$ is full rank, the only possible power allocation is

$$p = (D - S^{\text{target}}K)^{-1}S^{\text{target}}\sigma\mathbf{1}.$$

So the target SINR is achievable exactly when this p satisfies

$$0 \leq p_i \leq P^{\max}, \qquad i = 1, \ldots, n.$$

(b)

```
Largest feasible target SINR = 3.8
Power allocation = [0.09481765834932822, 0.04740882917466411, 0.05470249520153551]
```

```julia
using LinearAlgebra

G = [1   0.2 0.1;
     0.1 2   0.1;
     0.3 0.1 3]

sigma = 0.01
Pmax = 0.1
n = size(G, 1)

D = Diagonal(G)
K = G - D

best_S = nothing
best_p = nothing

for S in 2.0:0.1:4.0
    p = (D - S * K) \ (S * sigma * ones(n))
    feasible = all((p .>= 0) .& (p .<= Pmax))

    if feasible
        best_S = S
        best_p = p
    end
end
```

6 Interpolation with Rational Functions

(a)

For a fixed degree m, the interpolation condition is

$$y_i = \frac{a_0 + a_1 x_i + \cdots + a_m x_i^m}{1 + b_1 x_i + \cdots + b_m x_i^m}.$$

Multiplying by the denominator gives

$$a_0 + a_1 x_i + \cdots + a_m x_i^m - y_i b_1 x_i - \cdots - y_i b_m x_i^m = y_i.$$

This is linear in the unknown coefficients

$$\theta = \begin{bmatrix} a_0 & a_1 & \cdots & a_m & b_1 & \cdots & b_m \end{bmatrix}^T.$$

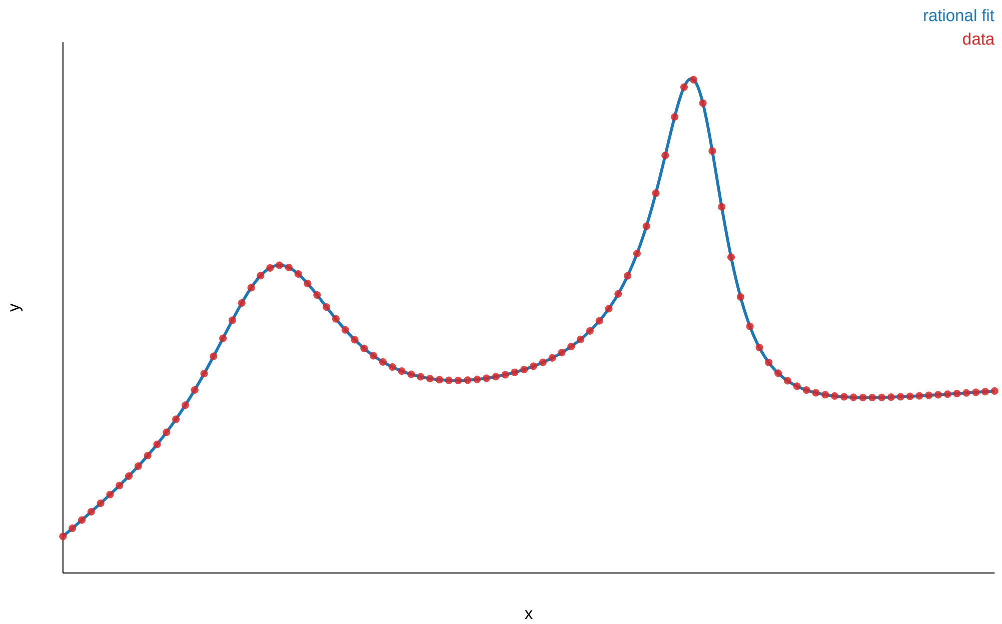
So for each $m = 0, 1, 2, \ldots$, we form the linear system

$$\begin{bmatrix} 1 & x_i & \cdots & x_i^m & -y_i x_i & \cdots & -y_i x_i^m \end{bmatrix} \theta = y_i, \qquad i = 1, \ldots, N.$$

The smallest degree m is the first value for which this system is consistent and the resulting rational function has degree m.

(b)

```
Smallest degree m = 5
a
   a_0 = 0.274181818181816
   a_1 = 1.02914438502676
   a_2 = 1.29064171123001
   a_3 = -5.87633689839592
   a_4 = -2.67379679144395
   a_5 = 6.68449197860985
b
   b_1 = -1.25133689839573
   b_2 = -6.51069518716569
   b_3 = 3.27540106951876
   b_4 = 17.3796791443848
   b_5 = 6.68449197860931
error = 4.219e-14
```




```
using LinearAlgebra
using JSON3
using Printf

data_file = joinpath(@__DIR__, "rational_interpolation_data.json")
data = JSON3.read(read(data_file, String))
x = Float64.(data.x.data)
y = Float64.(data.y.data)

function interpolation_matrix(x, y, m)
    numerator_terms = [x .^ j for j in 0:m]
    denominator_terms = [-(y .* (x .^ j)) for j in 1:m]
    return hcat(numerator_terms..., denominator_terms...)
end

function rational_value(x, a, b)
    numerator = sum(a[j + 1] * x^j for j in 0:(length(a) - 1))
    denominator = 1.0 + sum(b[j] * x^j for j in 1:length(b))
    return numerator / denominator
end

for m in 0:div(length(x) - 1, 2)
    A = interpolation_matrix(x, y, m)
    theta = A \ y
    residual = norm(A * theta - y, Inf)

    if residual <= 1e-8
        a = theta[1:(m + 1)]
        b = theta[(m + 2):end]
        yhat = [rational_value(xi, a, b) for xi in x]

        max_error = norm(yhat - y, Inf)
```

```
            break
      end
end
```